

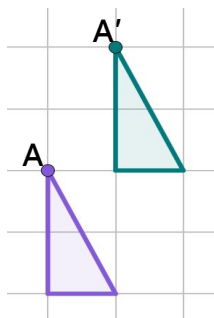


Describing a translation

Task A: Describing a translation

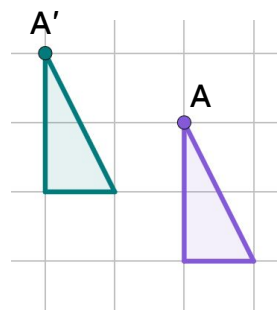
1) Complete the description for each image.

a)



The object has been translated
right by 1 squares and
up by 2 squares.

b)



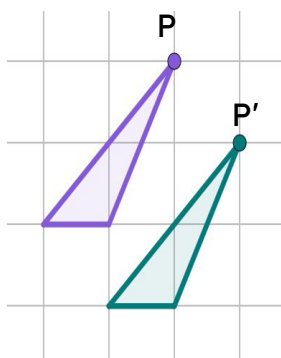
The object has been translated
left by 2 squares and
up by 1 squares.

2) Izzy has given an incomplete description of a translation.

The shape has moved across and down.



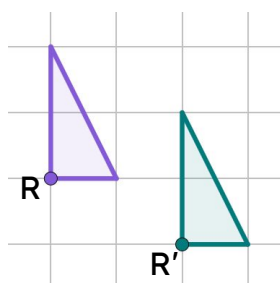
Write a full description for the translation shown.



The object has been translated
right by 1 square and *down* by 1 square.

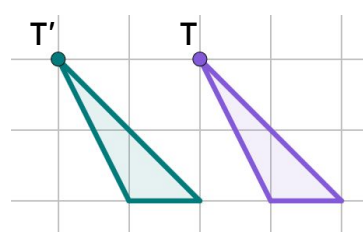
3) Write full descriptions for the translations.

a)



The object has been translated *right* by 2 squares and *down* by 1 square.

b)



The object has been translated *left* by 2 squares.



Task B: Vector notation

1) Match the description to the column vector.

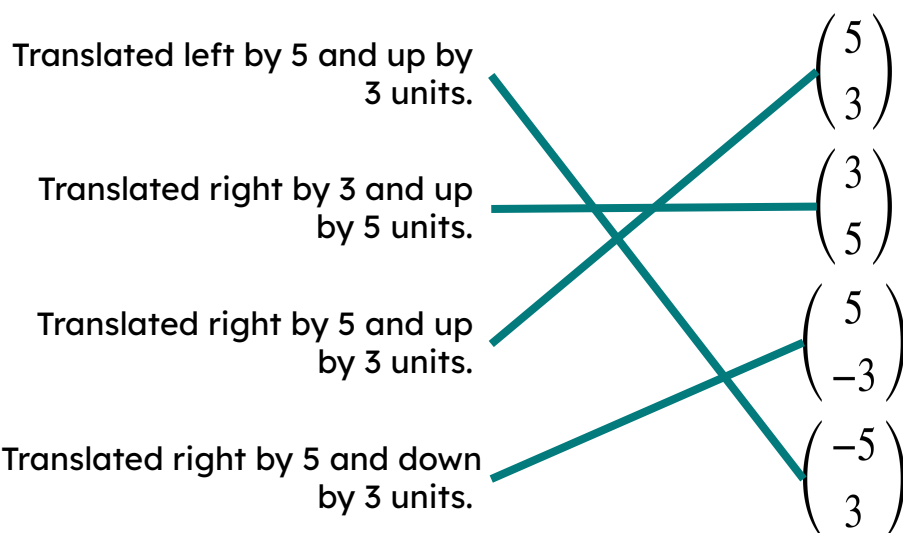
Translated left by 5 and up by 3 units.

Translated right by 3 and up by 5 units.

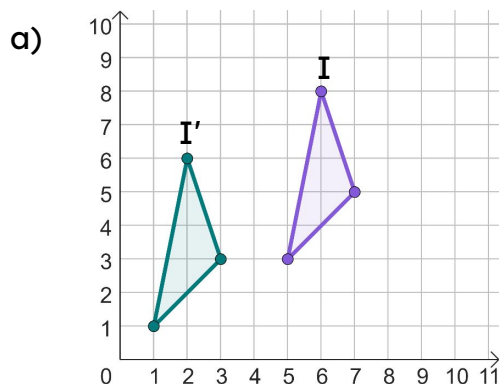
Translated right by 5 and up by 3 units.

Translated right by 5 and down by 3 units.

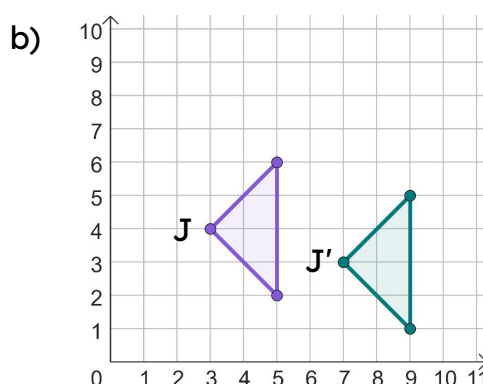
$\begin{pmatrix} 5 \\ 3 \end{pmatrix}$
 $\begin{pmatrix} 3 \\ 5 \end{pmatrix}$
 $\begin{pmatrix} 5 \\ -3 \end{pmatrix}$
 $\begin{pmatrix} -5 \\ 3 \end{pmatrix}$



2) Write a description using a column vector for each translation.



Translated by $\begin{pmatrix} -4 \\ -2 \end{pmatrix}$



Translated by $\begin{pmatrix} 4 \\ -1 \end{pmatrix}$

3) Given the coordinates of the object and image, describe the translation.

a)	Object coordinate	Image coordinate	b)	Object coordinate	Image coordinate
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(-2, 2)

(3, 1)

(2, 6)

(-1, 8)

(-2, -2)

(3, -3)

(4, 3)

(1, 5)

(1, -2)

(6, -3)

(5, 6)

(2, 8)

Translated by $\begin{pmatrix} 5 \\ -1 \end{pmatrix}$

Translated by $\begin{pmatrix} -3 \\ 2 \end{pmatrix}$

